

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report

```

```

iris = load_iris()
df = pd.DataFrame(iris.data, columns=iris.feature_names)
df['species'] = iris.target
df['species'] = df['species'].map({0: 'setosa', 1: 'versicolor', 2: 'virginica'})

print("Dataset Shape:", df.shape)
df.head()

```

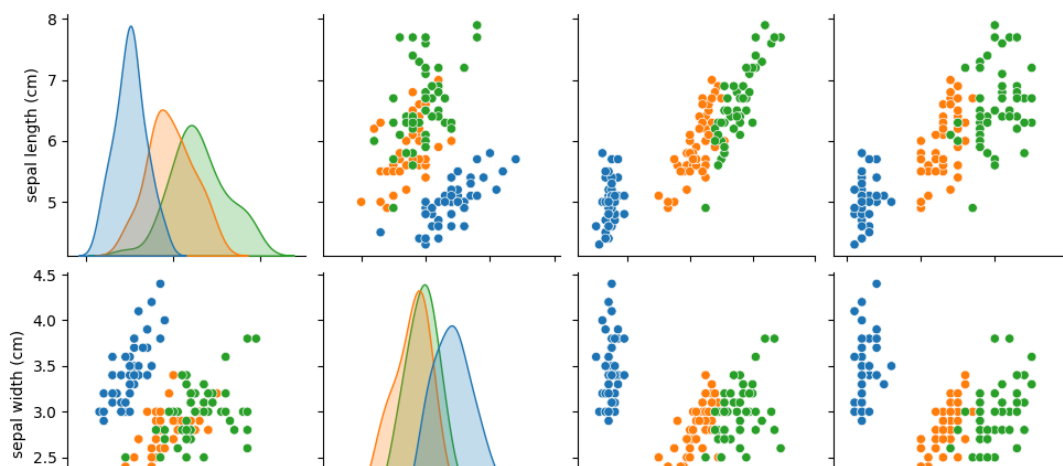
Dataset Shape: (150, 5)

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

```

sns.pairplot(df, hue='species')
plt.show()

```



```

X = iris.data
y = iris.target
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = LogisticRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred)*100, "%")
print("\nReport:\n", classification_report(y_test, y_pred))

```

Accuracy: 100.0 %

Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	10
1	1.00	1.00	1.00	9
2	1.00	1.00	1.00	11
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

```

print("Dataset Shape:", df.shape)
print("Total Rows:", df.shape[0])
print("Total Columns:", df.shape[1])

```

Dataset Shape: (150, 5)
Total Rows: 150
Total Columns: 5

```

iris = load_iris()
df = pd.DataFrame(iris.data, columns=iris.feature_names)
df['species'] = iris.target
df['species'] = df['species'].map({0: 'setosa', 1: 'versicolor', 2: 'virginica'})

print("Dataset Shape:", df.shape)
df.head()

```

Dataset Shape: (150, 5)

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

```
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Species ka count dekho
print(df['species'].value_counts())

# 2. Basic statistics
print("\nData ka summary:\n", df.describe())

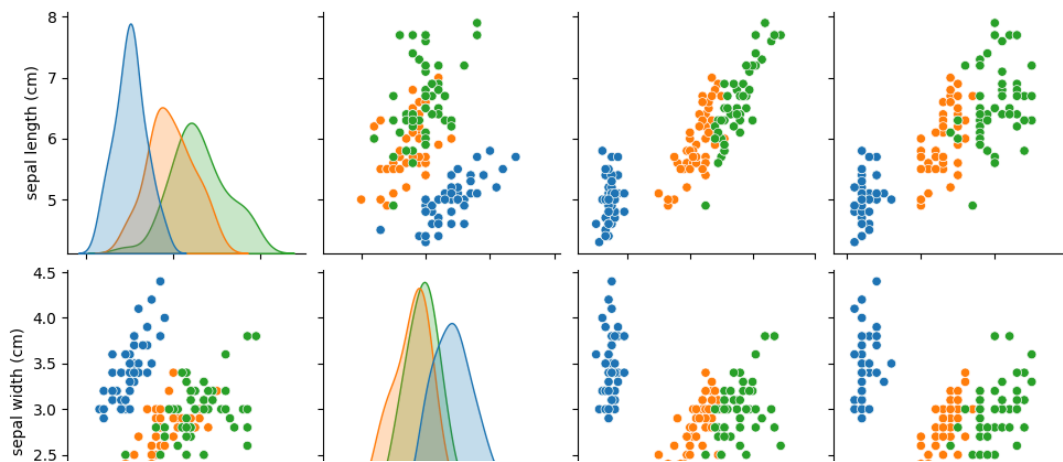
# 3. Sabse important graph - Pairplot
sns.pairplot(df, hue='species', height=2.5)
plt.suptitle('Iris Dataset - Flower Comparison', y=1.02)
plt.show()
```

```
species
setosa      50
versicolor 50
virginica   50
Name: count, dtype: int64
```

Data ka summary:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.198667
std	0.828066	0.435866	1.765298	0.469911
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.400000
50%	5.800000	3.000000	4.350000	1.000000
75%	6.400000	3.300000	5.100000	1.300000
max	7.900000	4.400000	6.900000	2.000000

Iris Dataset - Flower Comparison



```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

# 1. Data ko train aur test me split
X = iris.data
y = iris.target
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# 2. Data ko scale karna - important hai
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# 3. Model train karo
model = LogisticRegression()
model.fit(X_train, y_train)

# 4. Prediction aur Accuracy
y_pred = model.predict(X_test)
acc = accuracy_score(y_test, y_pred)
```

